

Effect of Switching Frequency on Efficiency in Wide Bandgap Based Drive Systems

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Abstract—This paper investigates the effect of switching frequency on the losses occurring in the inverter and motor of an electric vehicle throughout the WLTC. A combination of analytical and numerical methods are used to model the conduction and switching losses in the inverter, and the copper and iron losses in the motor. Iron losses are derived from a Steinmetz model. The Tesla Model 3 is used as a reference vehicle to size the drive system. For the selected drive system parameters, the switching frequencies that minimise the overall system losses for Si, SiC, and GaN based inverters are found. The reduction in switching losses of SiC and GaN over Si result in an increase in the optimum switching frequency, from 4 kHz to 11 kHz and 20 kHz respectively.

Index Terms—Switching Frequency, Silicon Carbide, Gallium Nitride, Output Current Ripple, WLTC

NOMENCLATURE

ΔB	Peak-to-peak flux density change
Δi_{inv}	Inverter ripple current RMS
Δi_m	Motor ripple current RMS
κ	Gearbox ratio
$\cos \phi$	Power factor
ω_e	Electrical angular frequency
ω_m	Motor speed
ρ	Air density
a	Vehicle acceleration
A_f	Frontal area
B	Flux density
C_D	Drag coefficient
C_{rr}	Rolling resistance coefficient
$E_{sw}(\cdot)$	Switching energy loss function
f_{sw}	Switching frequency
i_d	Direct current
i_{inv}	Inverter fundamental current RMS
i_{mean}	Inverter mean absolute current
i_m	Motor fundamental current RMS
i_q	Quadrature current
k	Drive cycle data sample

K_e	Motor constant
k_i, α, β	Steinmetz coefficients
L_q	Direct inductance
L_q	Quadrature inductance
M	Modulation index
m	Vehicle and driver mass
M_{iron}	Iron mass
n	Number of transistors in parallel
p	Pole pairs
P_{cond}	Inverter Conduction loss
P_{cu}	Motor Copper loss
P_{iron}	Motor iron loss
P_{sw}	Inverter switching loss
$R_{ds(on)}$	Transistor on-state resistance
R_{fund}	Winding fundamental frequency resistance
R_{rip}	Winding switching frequency resistance
R_s	Winding DC resistance
r_T	Tire radius
r_{ce}	Si IGBT forward resistance
r_F	Si IGBT diode resistance
T_c	Device case temperature
T_J	Device junction temperature
T_m	Motor torque
v	Vehicle speed
v_{bat}	Battery voltage
v_{inv}	Inverter fundamental phase voltage RMS
v_{ll}	Inverter line-to-line voltage RMS
v_m	Motor fundamental phase voltage RMS
V_{ce0}	Si IGBT forward voltage drop
V_{F0}	Si IGBT diode voltage drop

I. INTRODUCTION

The range of electric vehicles can be improved by increasing drive system efficiency. In this paper we define the drive system is defined as the inverter and motor. Changing the switching frequency of the inverter changes the losses in the drive system; typically increasing the switching frequency tends to reduce losses in the motor [1] due to reduced ripple current, but increases losses in the inverter. Therefore, there is an optimum switching frequency at which the losses in the inverter and motor together are minimum. Previous studies have shown that the optimisation of the switching frequency

This research was funded in whole, or in part, by the UK Engineering and Physical Sciences Research Council (EP/W524311/1) and YASA Limited. For the purpose of Open Access, the author has applied a CC BY public copyright licence to any Author Accepted Manuscript version arising from this submission. No new data was created during the study.

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of a Si IGBT-based inverter can improve the efficiency of the overall drive system by 7.5% [2] .

Wide bandgap (WBG) power devices, such as silicon carbide (SiC) MOSFETs and gallium nitride (GaN) HEMTs can offer lower switching losses when compared to traditional Si IGBTs [3]. [3] reported a $\times 4.4$ reduction in the switching losses of SiC MOSFET vs. Si IGBT, and a $\times 14$ reduction in the switching losses of GaN HEMT vs. Si IGBT. Newer SiC inverters tend to use a higher switching frequency than older Si inverters, e.g. the Tesla Model 3 at 12 kHz verses the Nissan Leaf at 5 kHz [4].

The efficiency of a drive system varies considerably depending on the operating point (defined as the motor torque and speed) [5], [6]. Therefore it is important to use a representative drive cycle to capture the average efficiency under real world conditions. The use of a drive cycle is a common method in electric vehicle optimisation studies [7].

In this paper, the inverter switching frequency is varied and the losses induced by the fundamental and ripple components of the current in the inverter and motor are modelled to determine the total losses in the drive system. The inverter is a two-level, three-phase, voltage source inverter, using a symmetric Space Vector Pulse Width Modulation scheme. Si IGBT, SiC MOSFET and GaN HEMT switches are considered in this paper. The motor is an Interior Permanent Magnet Synchronous Machine (IPMSM). Fundamental and ripple losses in the inverter and motor are computed analytically, using parameters drawn from various sources.

II. METHOD

Fig. 1 shows an electrical schematic of the battery and drive system, and indicates the losses modelled in this study.

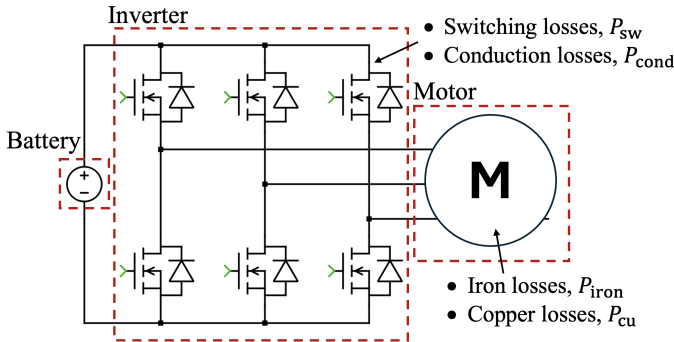


Fig. 1. Electrical schematic of the drive system, including location of the losses modelled.

Fig. 2 illustrates the process to derive the inverter phase voltage and current from the WLTC data. It also shows how the inverter phase voltage and current are used to compute the drive system losses. For calculating the operating points, and then the loss modelling, the system is assumed to be quasi-static: e.g. $v_d[k]$ is the steady state value of v_d for a given $v[k]$ and $a[k]$

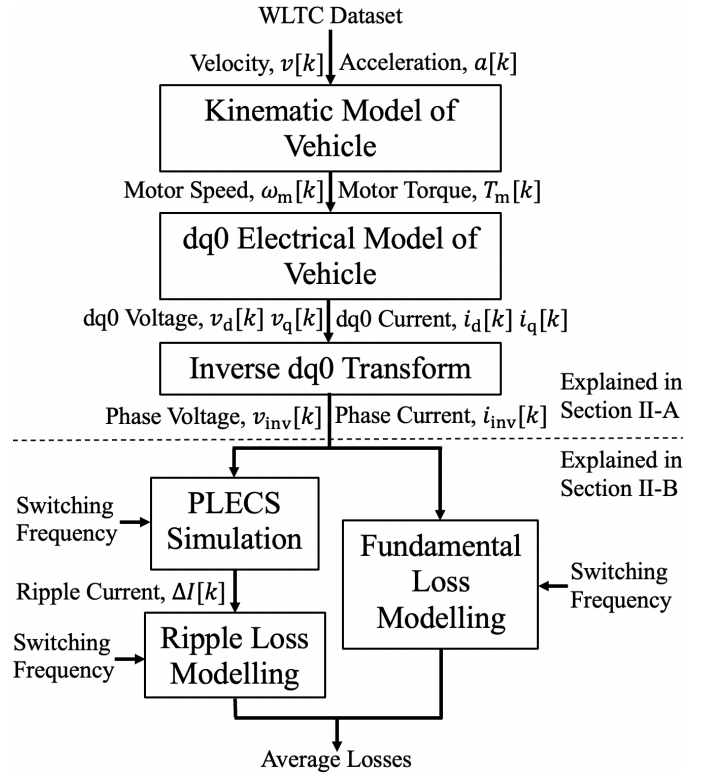


Fig. 2. Block diagram detailing the process of using the WLTC to acquire the set of inverter phase voltages and currents.

A. Drive Cycle

In [8], a number of drive cycles were compared, including the Worldwide Harmonised Light Vehicles Test Cycle (WLTC) [9], a drive cycle proposed by the United Nations Economic Commission for Europe. It was found that the WLTC most closely modelled conditions for on-the-road driving. For this reason, the WLTC is chosen as the drive cycle for this study.

The WLTC is a series of 1800 data points at 1 second intervals for the speed of a vehicle over a journey of 30 minutes. These data points are used to derive the operating points and drive system losses, also at 1 second intervals.

For most of the properties of the vehicle, the Rear-Wheel Drive Tesla model 3 is used as the reference case (a representative average passenger electric vehicle). Its properties are widely available [10]–[12]. Table I details the properties of the vehicle, motor, and inverter.

Firstly, the velocity, and acceleration of the vehicle is used to calculate the required tire force:

$$F_T = \frac{1}{2} \rho v^2 A_f C_D + C_{rr} mg + ma \quad (1)$$

The motor torque and angular speed are calculated as:

$$T_m = \frac{F_T r_T}{\kappa} ; \quad \omega_m = \frac{v \kappa}{r_T} \quad (2)$$

Fig. 3 shows the motor torque and speed verses time over the drive cycle.

TABLE I
VEHICLE, MOTOR, AND INVERTER PROPERTIES

	Property	Value	Ref.
Vehicle	Vehicle and driver mass, m	1840 kg	[10]
	Drag coefficient, C_D	0.23	[10]
	Frontal area, A_f	2.22 m ²	[10]
	Air density, ρ	1.2 kg/m ³	
	Tire radius, r_T	33.4 cm	[10]
	Rolling resistance coefficient, C_{rr}	0.01	
	Gearbox Ratio, κ	1:9	[10]
Motor ^a	Motor constant, K_e	0.23 Vs/rad	[11]
	Stator DC resistance, R_s	4.75 m Ω	[11]
	Quadrature inductance, L_q	603 μ H	[12]
	Direct inductance, L_d	225 μ H	[12]
	Pole pairs, p	3	[12]
	Current flux density ratio, γ	4 mT/A	[11]
	Iron mass, M_{iron}	25 kg ^b	[12]
	Iron material	M270-35A	[12]
Inverter	Voltage rating	400 V	[12]
	Current rating	720 A	
	Case temperature for inverter sizing, T_c	100 °C	
	Junction temperature for device properties, T_J	100 °C	

^aStar connected. ^bIron mass estimated from total motor mass.

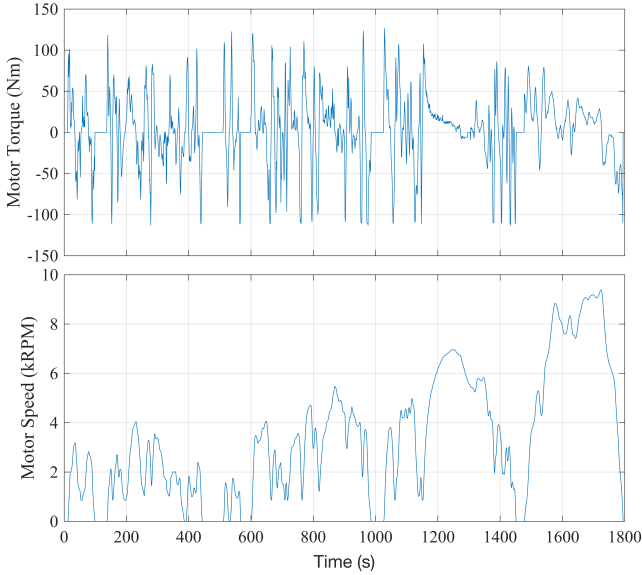


Fig. 3. Motor torque and speed verses time.

At the points where the motor torque is negative, regenerative braking is assumed to provide all negative torque, as opposed to conventional braking.

Once the motor torques and rotational speeds are calculated, a representation of the motor in the dq0 reference frame is used to derive the motor d and q axis currents and voltages. This is done by selecting the pair of i_q and i_d , which minimise the total motor phase current RMS whilst delivering the required motor torque, and ensuring that the motor phase voltage does not exceed the limit imposed by the battery voltage. This is expressed as:

$$\begin{aligned} \min \quad & \sqrt{i_q^2 + i_d^2} \\ \text{s.t.} \quad & T_m = \frac{3}{2} K_e i_q + \frac{3}{2} p (L_d - L_q) i_q i_d \\ & v_m \leq \frac{V_{bat}}{\sqrt{6}} \end{aligned} \quad (3)$$

The motor phase voltage RMS is limited to less than $V_{bat}/\sqrt{6}$, because when using SVPWM, the maximum RMS of the line-to-line voltage from an inverter is equal to $V_{bat}/\sqrt{2}$. The RMS of the motor phase voltage is:

$$v_m = \frac{1}{\sqrt{2}} \sqrt{v_q^2 + v_d^2} \quad (4)$$

The motor voltages in the dq0 reference frame are:

$$v_q = R_s i_q + \omega_e L_d i_d + K_e \omega_m ; \quad v_d = R_s i_d - \omega_e L_q i_q \quad (5)$$

The electrical angular frequency is computed using $\omega_e = p\omega_m$.

Fig. 4 shows motor voltage and current in the dq0 reference frame verses time over the drive cycle.

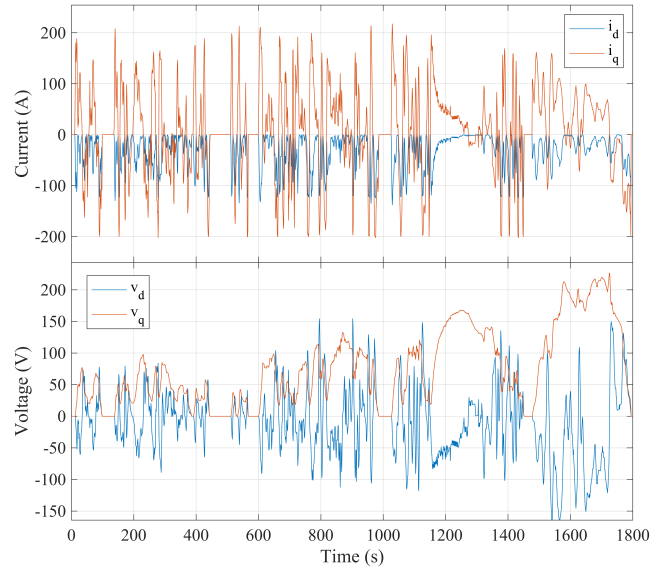


Fig. 4. Motor voltage and current in the dq0 reference frame verses time.

Once the voltages and currents in the dq0 reference frame are known, the power-invariant inverse park and clark transforms can be used to find the required motor phase current and voltage in the abc reference frame.

The inverter phase current equals the motor phase current, as it is a star connected motor. The motor phase voltage can be used to find the line-to-line voltage, which in turn can be used to find the phase inverter voltage. This is detailed in (6).

$$v_{inv} = \frac{v_{ll}}{\sqrt{3}} = v_m ; \quad i_{inv} = i_m \quad (6)$$

Fig. 5 shows the inverter phase voltage RMS and current RMS, throughout the whole of the drive cycle. These time waveforms are used as the operating points for measuring the efficiency of the drive system, detailing the required phase voltage and phase current in one second intervals.

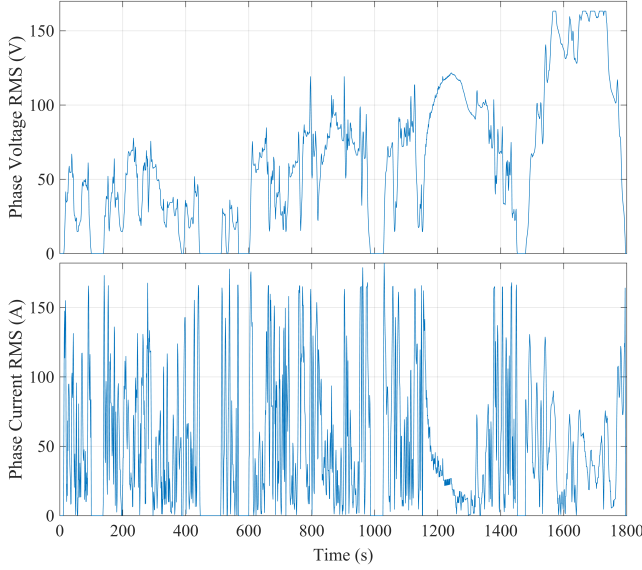


Fig. 5. Inverter phase voltage RMS and phase current RMS verses time.

B. Loss Modelling

PLECS is used to calculate the ripple current by modeling two drive systems: one with the PWM inverter, and one with an average model of the inverter. The difference between the two currents calculated in the models is the ripple current. This is a similar approach to [13] where SPICE is used to compute a ripple current from the difference of a PWM inverter and average model of the inverter.

To compute switching losses, switching energy data from the selected switches data sheet is used with an interpolating function to find the switching losses at each point in the drive cycle. The switching losses are a linear function of the current for the range of currents expected, and so the mean switching energy is the switching energy of the mean current. Equation (7) shows how the inverter mean absolute current relates to the RMS inverter phase current, calculated before.

$$i_{\text{mean}} = \frac{2\sqrt{2}}{\pi} i_{\text{inv}} \quad (7)$$

The ripple and fundamental currents are assumed to be orthogonal and therefore the ripple current does not contribute to the switching losses, as it is zero mean. The switching losses area:

$$P_{\text{sw}} = 6E_{\text{sw}} \left(\frac{i_{\text{mean}}}{n} \right) f_{\text{sw}} n \quad (8)$$

$E_{\text{sw}}(\cdot)$ is the sum of the turn-on and turn-off energy of one transistor, for a given mean absolute current conducted.

To calculate the conduction losses in the inverter, (9) is used for MOSFETs and HEMTs. For the case of the MOSFET, when considering the maximum current that is conducted by a transistor in the inverter throughout the drive cycle, and their $R_{\text{ds(on)}}$, it can be concluded that the voltage drop across any one of the transistors never exceeds the body diode forward voltage, meaning during reverse conduction the body diode conducts none of the current. HEMTs have no intrinsic body diode, meaning reverse conduction occurs through the same channel as forward conduction does. This means for the task of calculating the conduction losses, it is assumed that one transistor channel per half-bridge always conducts the phase current (the effects of dead time are neglected in this study), and so:

$$P_{\text{cond}} = 3 \left(\left(\frac{i_{\text{inv}}}{n} \right)^2 + \left(\frac{\Delta i_{\text{inv}}}{n} \right)^2 \right) R_{\text{ds(on)}} n \quad (9)$$

For the IGBTs [14]:

$$\begin{aligned} P_{\text{cond}} = 6n & \left(\frac{V_{\text{ce0}}}{2\pi} \frac{\sqrt{2}i_{\text{inv}}}{n} \left(1 + \frac{\pi}{4} M \cos \phi \right) \right. \\ & + \frac{r_{\text{ce}}}{2\pi} \left(\frac{\sqrt{2}i_{\text{inv}}}{n} \right)^2 \left(\frac{\pi}{4} + \frac{2}{3} M \cos \phi \right) \\ & + \frac{V_{\text{F0}}}{2\pi} \frac{\sqrt{2}i_{\text{inv}}}{n} \left(1 - \frac{\pi}{4} M \cos \phi \right) \\ & \left. + \frac{r_{\text{F}}}{2\pi} \left(\frac{\sqrt{2}i_{\text{inv}}}{n} \right)^2 \left(\frac{\pi}{4} - \frac{2}{3} M \cos \phi \right) \right) \end{aligned} \quad (10)$$

The power factor is calculated as:

$$\cos \phi = \cos \left(\arctan \left(\frac{v_{\text{q}} i_{\text{d}} - v_{\text{d}} i_{\text{q}}}{v_{\text{d}} i_{\text{d}} + v_{\text{q}} i_{\text{q}}} \right) \right) \quad (11)$$

Similarly to conduction losses in the inverter, copper losses in the motor are modelled by calculating the RMS value of the fundamental and ripple currents. The key difference is that because the copper in the motor is in the form of thin wires, the skin and proximity effect can alter the apparent resistance of the copper. Due to this, the contribution of the copper losses from the fundamental and ripple currents both have modified resistances that consider the skin and proximity effect at their respective frequencies. This is done by modifying R_{s} using the Dowell equation [15] to calculate R_{fund} and R_{rip} , and so:

$$P_{\text{cu}} = 3 (i_{\text{m}}^2 R_{\text{fund}} + \Delta i_{\text{m}}^2 R_{\text{rip}}) \quad (12)$$

While the flux density magnitude is not in practice equal throughout the motor iron (certain areas such as the tips of the stator teeth and inside corners of core material, typically have a higher flux density than the surrounding material), to fully model the flux distribution would require extensive use of finite element analysis, which is computationally expensive.

For this study then, the flux density distribution is assumed to be equal throughout the core material of the motor.

Iron losses resulting from the ripple current are modelled using the improved General Steinmetz Equation (iGSE) [16]. The iGSE was derived for modelling iron losses in systems that experience non-sinusoidal flux density waveforms. While there are other adjusted Steinmetz equations, such as the modified Steinmetz equation and general Steinmetz equation, the iGSE has been shown to characterise the iron losses in non-sinusoidal waveforms more accurately than these other methods [16].

When the ripple current waveforms are computed such that the RMS value can be calculated, the iGSE is also used to find the iron losses due to the ripple current in the motor. The flux density in the motor is modelled to be proportional to the phase current magnitude, as the motor assumed to have a constant inductance (i.e., the iron is assumed to have constant permeability with flux density). The ratio of the current to flux density (γ) is calculated from the FEA simulation in [11]. The relationship between the motor ripple current and flux density is:

$$B = \sqrt{2}\gamma\Delta i_m \quad (13)$$

Once the ripple flux density is calculated from a given ripple current, the average iron loss due to the ripple current is:

$$P_{\text{iron}} = 3M_{\text{iron}} \frac{k_i}{T} \int_0^T \left| \frac{dB}{dT} \right|^\alpha |\Delta B|^{\beta-\alpha} dt \quad (14)$$

Fig. 6 is an example of a small section of the ripple current waveform calculated in PLECS. The solid black vertical lines indicate the beginning and end of every subloop on which the iGSE equation is applied to, as detailed in [16].

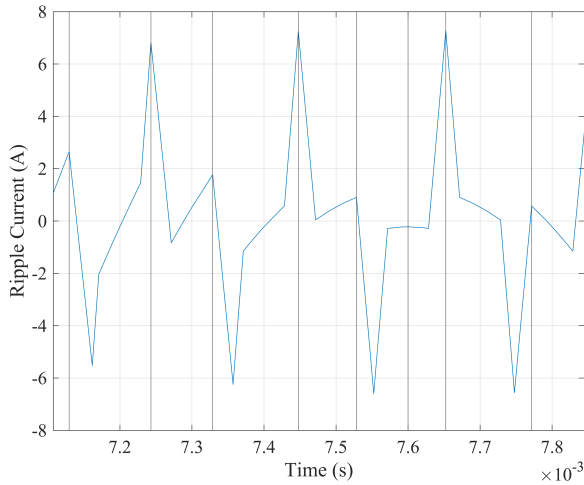


Fig. 6. A section of the ripple current with solid black vertical lines indicating the start and end of every subloop on which the iGSE equation is applied to.

For the fundamental iron losses, the basic Steinmetz equation is used, as the fundamental current is sinusoidal. The accuracy of computing the total iron losses is reduced when using the method of summing the individually calculated

losses from the fundamental and ripple currents, due to the non-linear nature of iron losses. However, previous studies [2], [17] have used a linear summation of iron loss harmonics method to calculate iron losses, with good agreement between theoretical and experimental results.

III. RESULTS AND DISCUSSION

Table II lists the three selected transistors used in this study. They were selected based on their advertised best-in-class performance. Included in the table is the current rating of an individual transistor, and the number of transistors in parallel required to achieve the rated inverter current. Also included are the values used for calculating conduction losses, derived from the relevant datasheets. For sizing the number of parallel devices in the inverter, the manufacturers rated continuous current at the case temperature specified in Table I is used. The device properties used to compute the switching and conduction losses are at the junction temperature also specified in Table I. The use of a specified case and junction temperature for different parts of the analysis removes the need for a thermal model of the devices and their cooling system.

TABLE II
SELECTED 650 V TRANSISTORS AND CHARACTERISTICS

	Si IGBT	SiC MOSFET	GaN HEMT
Model	IKZA50N65EH7	C3M0025065K	SGT65R65AL
Producer	Infineon	Wolfspeed	STMicroelectronics
Rated current @ T_c	76 A	70 A	25 A
n @ T_c	9.5	10.3	28.8
$R_{ds(on)}$ @ T_J	-	27.5 m Ω	84 m Ω
V_{ce0} @ T_J	0.47 V	-	-
r_{ce} @ T_J	27.4 m Ω	-	-
V_{F0} @ T_J	0.54 V	-	-
r_F @ T_J	26.5 m Ω	-	-

Figure 7 plots the individual and total drive system losses for Si, SiC, and GaN switches, averaged over the drive cycle. The minimum average total drive system losses for each system are marked with the red stars; Si: 565 W at 4 kHz, SiC: 401 W at 11 kHz, and GaN: 370 W at 20 kHz.

As can be seen in Fig. 7, the two losses which compete to create a minimum are switching losses and iron losses. Copper and conduction losses have only a small influence on the location and value of the minima. Conduction losses are particularly small due to the large number of transistors in parallel to achieve the required current rating, which is $\times 4.5$ greater than the peak RMS current required by the drive cycle, meaning that the apparent $R_{ds(on)}$ or r_{ce0} of the legs in the inverter are small. It can also be seen that the ripple current component of the copper and conduction losses have little effect on the total losses.

As expected, the lower switching losses in SiC and GaN transistors results in a higher optimum switching frequency, as well as a lower total power loss, when compared to Si: The optimum switching frequency for Si and SiC respectively correlate well with values for electric vehicles using those transistor types.

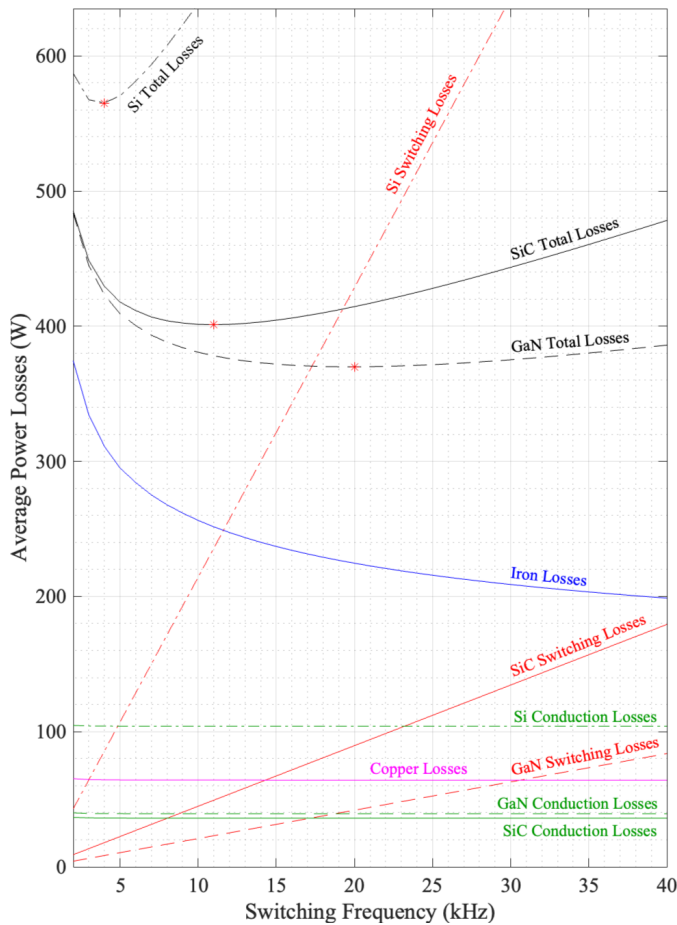


Fig. 7. Plot of the average individual and total power losses in the drive system verses switching frequency, for Si, SiC, and GaN switches. Note the red stars, indicating points of minimum average power losses.

In the case of GaN transistors the total power loss minimum is very weak, with less than a 10% variation between the switching frequencies of 6 kHz and 40 kHz.

IV. CONCLUSION

In this paper, the effect of transistor type and switching frequency on the efficiency of a drive system has been investigated. The method computes the drive system losses throughout a drive cycle for a range of switching frequencies, calculating the expected average losses in the inverter and motor.

Optimum switching frequencies for Si, SiC, and GaN switches have been found, and reinforce the improvement that is achieved by replacing Si IGBTs with SiC MOSFETs, and indicate the improvement that could be achieved by replacing SiC MOSFETs with GaN HEMTs. Future work should focus on improving the motor iron loss model to better reflect the actual flux distribution in a motor.

ACKNOWLEDGMENT

The authors would like to acknowledge Tony Webster, Paul Spendley, Raul Martins, and Rajesh Kudikala, of YASA Motors, for their input to this project.

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